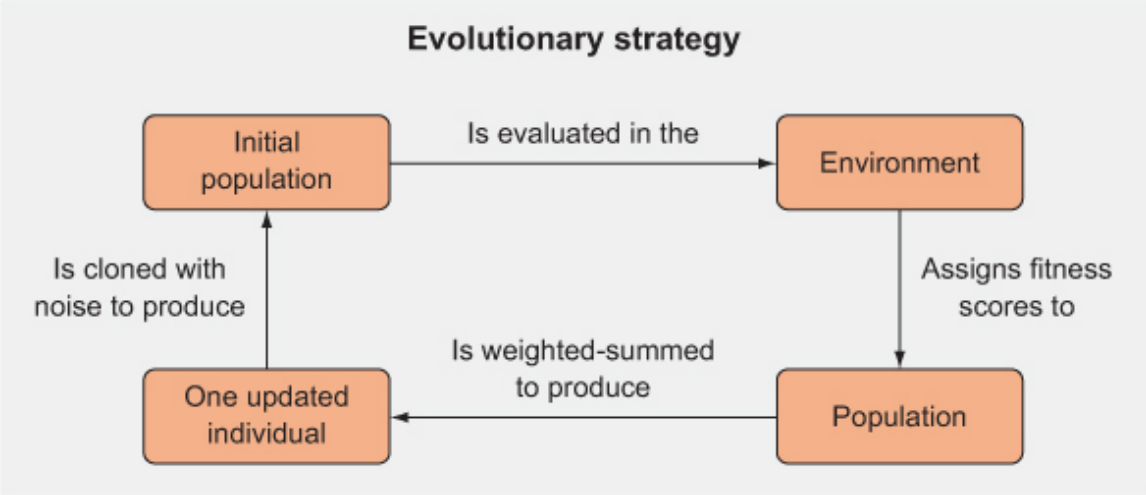
There's another class of evolutionary algorithms confusingly termed *evolutionary strategies* (ES), which employ a less biologically accurate form of evolution, as illustrated in figure 6.10.

Figure 6.10. In an evolutionary strategy we create a population of individuals by repeatedly adding a small amount of random noise to a parent individual to generate multiple variants of the parent. We then assign fitness scores to each variant by testing them in the environment, and then we get a new parent by taking a weighted sum of all the variants.



If we're training a neural network with an ES algorithm, we start with a single parameter vector θ_t, sample a bunch of noise vectors of equal size (usually from a Gaussian distribution), such as $e_i \sim N(\mu,\sigma)$, where N is a Gaussian distribution with mean vector μ and standard deviation σ. We then create a population of parameter vectors that are mutated versions of θ_t by taking $\theta'_i = \theta + e_i$. We test each of these mutated parameter vectors in the environment and assign them fitness scores based on their performance in the environment. Lastly, we get an updated parameter vector by taking a weighted sum of each of the mutated vectors, where the weights are proportional to their fitness scores (figure 6.11).

Figure 6.11. In an evolutionary strategy, at each time step we get an updated parameter vector by taking the old parameter vector and adding it to a weighted sum of the noise vectors, where the weights are proportional to the fitness scores.